



Collaborative VR environment for supporting food culture learning

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ABSTRACT

In recent years, cultural learning has required the design of experiential and reflective environments that go beyond mere knowledge acquisition to support the understanding and internalization of cultural values. This study constructed a collaborative VR learning environment focused on food culture, enabling learners to actively interpret and engage in the meaning-making of cultural values in a virtual space. The system consists of scenarios based on three themes: ingredients, tableware, and cuisine. It offers experiences that engage learners with cultural elements through exploration, observation, and cooking. Through voice and non-verbal gestures via avatars, learners shared their insights and emotions, deepening their learning. A comparative experiment showed that participants in the collaborative learning condition significantly outperformed those in the individual learning condition in terms of recognition of cultural values, knowledge acquisition, learning motivation, and cultural awareness. Observations also revealed the natural formation of joint attention through gestures such as pointing and shared gaze, which facilitated reflective meaning-making. Based on these findings, four design recommendations were derived: (1) supporting meaning-making through experience and dialogue, (2) supporting collaboration through non-verbal interaction, (3) designing immersive experiences that enhance intrinsic motivation, and (4) creating scenarios and tasks that encourage reflection.

1. Introduction

Food culture fosters cultural identity and social connection through multifaceted practices, including the selection of ingredients, cooking methods, the use of tableware, and dining etiquette [1,2]. It also serves as a medium for intercultural understanding and international exchange, offering significant educational value [3]. Cultural education should nurture understanding and respect for cultural values, thereby fostering more inclusive, agile, and resilient societies [4]. However, due to the abstract and multilayered nature of cultural values, it is difficult for learners to deeply understand the meanings and emotions behind them through descriptive teaching materials alone. UNESCO emphasizes that experiential and participatory learning is essential for cultural understanding [4]. Ham similarly notes that learners are not passive recipients of information, but participants who construct meaning through experience [5].

Given this background, cultural learning environments must be designed to encourage sensory and physical engagement, thereby facilitating meaning-making. For example, Muntean et al. [6] proposed an interactive museum exhibit using tangible interfaces that allow visitors to engage with intangible cultural values through active interaction. Moreover, VR (Virtual Reality) has gained attention as a means to present cultural situations that are difficult to reproduce in the real

world. By integrating visual, auditory, and spatial senses, VR provides immersive experiences that allow learners to actively engage with cultural contexts. For instance, Puig et al. [7] demonstrated that VR-based museum experiences, which combine reconstructed cultural scenarios with learner behavior, promote meaning-making and reflection.

Although research on VR-supported cultural learning remains limited, several studies have demonstrated its educational effectiveness. For example, Li et al. [8] reported that a VR learning environment focused on tea culture enhanced learners' empathy and motivation. Puig et al. [7] also showed that supplementing archaeological museum exhibitions with VR deepened understanding of cultural heritage. In fields such as STEM education, the use of VR has been shown to improve learning outcomes [9–11], and the educational effectiveness of immersive environments has been acknowledged across diverse disciplines. However, most existing studies focus on individual learning. They rarely address design strategies that support deep understanding and meaning-making of ambiguous and subjective cultural values through dialogue and collaboration.

To understand cultural values, it is essential to share perspectives and co-construct meaning through dialogue with others. Accordingly, collaborative learning has gained attention as a framework for facilitating this process. Stahl [12] views knowledge construction not

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as an accumulation of individual understanding, but as a process of meaning generation by groups. Koschmann [13] similarly defines learning as “meaning-making”. Dillenbourg [14] outlines key components of collaborative learning, including interdependence and negotiated dialogue. Furthermore, reflection is considered critical in collaborative learning. Boud et al. [15] argue that reflection is essential for engaging in meaning-making from experience and is often stimulated through interaction with others. Price et al. [16] also point out that linguistic processes such as opinion sharing and reinterpretation promote reflection.

Building on these perspectives, this study proposes a collaborative VR learning environment that integrates scenario-based cultural learning tasks. In this system, learners explore cultural scenarios in a shared virtual space and engage in dialogue using voice and gestures, enabling mutual exchange of perspectives and discoveries. The term “scenario” refers to specific contextual settings related to Japanese food culture, while “task” refers to the specific activities or challenges learners undertake within these scenarios. For example, through a series of activities such as collecting ingredients, appreciating tableware, and experiencing cooking, learners are encouraged to encounter cultural values and construct meaning collaboratively.

To investigate how such a learning environment facilitates learners’ understanding of cultural values and their interactions with others, the following two research questions are defined:

- **RQ1:** “How can scenario-based VR experiences facilitate learners’ understanding of cultural values in Japanese food culture through collaborative learning, without direct instruction?”
- **RQ2:** “What is the impact of collaborative learning, compared to individual learning, on learning outcomes and motivation in VR cultural learning environments?”

This study aims to address the above questions by examining the impact of collaborative learning in immersive VR environments on learners’ reflection and understanding of cultural values, as well as the effectiveness of such environments in terms of learning outcomes and motivation.

2. Related works

This section reviews previous studies related to supporting the understanding of cultural values through collaborative learning in VR environments. It examines the nature of cultural learning, meaning-making regarding cultural values, and educational practices utilizing VR and collaborative learning, thereby positioning the contribution of this study.

2.1. Cultural learning

2.1.1. Classifications of cultural learning

Cultural learning refers to the process of acquiring knowledge and values related to cultural practices such as history, art, language, and customs. It takes diverse forms depending on the learner’s objectives and interests. In particular, who the learners are significantly influences the motivation, learning approach, and design requirements of the learning environment.

From this perspective, research on cultural learning often classifies learners into two broad types. One is the “formal learner”, who learns based on educational obligations such as school education or vocational training. The other is the “informal learner”, who engages in learning voluntarily based on personal interest or curiosity. According to Ham [5], these two types of learners are characterized as follows.

Formal learners, such as professionals, researchers, and students, are those who aim to acquire systematic knowledge. They are expected to acquire as much information about culture as possible, and their learning outcomes are largely dependent on the quantity of knowledge

acquired. In contrast, informal learners begin learning based on what they see, hear, read, or experience, driven by their own interest. They can freely start or stop learning, and their continued engagement depends on how attractive and interesting they find the content.

This study emphasizes the importance of learners initiating learning from their own interests and meaning-making of cultural values through experiences. Therefore, it primarily targets informal learners, rather than those who engage in learning based on clearly defined goals or evaluation criteria like formal learners. While formal learning is supported by institutional frameworks and mainly aims at knowledge acquisition, informal learning positions the learner as the agent who constructs the meaning and value of learning.

Given the characteristics of informal learners, we focus on identifying the types of information they find engaging and the aspects they perceive as valuable. To this end, the study employs the Intrinsic Motivation Inventory (IMI) [17] to evaluate various dimensions of their intrinsic motivation. The IMI has been validated for its validity and reliability [18]. It includes subscales such as *Interest/Enjoyment* and *Value/Usefulness*, and has been widely used to assess learners’ intrinsic motivation.

Regarding the goals of cultural learning, Ham [5] believes that for cultural learning that leads to cultural awareness and protection, the main aim is not to teach but to provoke the learners into thinking about and engaging in meaning-making around cultural values. He also suggests that this type of cultural learning setup is suitable for informal learners. Lord [19] describes cultural learning in museums as an informal, voluntary, and effective “transformative experience”. Learning is effective when it affects visitors’ attitudes, interests, appreciation, beliefs, or values of culture. Ibrahim et al. [20] define cultural learning as informal learning that leads to historical and cultural awareness among learners. They suggest that the outcome of cultural learning is not merely mastery of facts or the amount of information that learners can recall academically. Instead, cultural learning should be able to provide a setting that encourages learners to think, discover, and make meaningful connections with the heritage by themselves.

In this way, learning outcomes in cultural learning are not formed simply by absorbing information, but rather through meaning-making that is rooted in learners’ own experiences and values. In this study, we evaluate learning outcomes of cultural learning based on the following three aspects: cultural knowledge acquisition, recognition of cultural values, and cultural awareness in life developed through experience. What is referred to as “cultural values” here are the meanings, beliefs, and behavioral patterns embedded in everyday life. The key is whether learners can recognize these values and reinterpret them through experience.

2.1.2. Cultural values and meaning-making

In cultural learning, it is not merely the acquisition of knowledge that matters, but rather how learners experience cultural values and engage in meaning-making around them. Cultural values are deeply embedded in people’s daily lives, beliefs, and behavioral patterns, and are shaped and shared within specific regional, historical, and social contexts. For example, concrete practices such as dining etiquette, the significance of festivals, and everyday customs inherently reflect the unique values of a given culture.

However, merely receiving such values as information is insufficient. Learners are required to reflect on the meanings behind these practices and reinterpret them in relation to their own experiences. In this sense, understanding cultural values is less about knowledge acquisition and more akin to a process of meaning-making through lived experience. This view aligns with Oyserman’s concept of culture as a “cultural mindset” [21]—not a static system of knowledge, but something that is activated in particular contexts and influences individuals’ behaviors and judgments.

From this perspective, the design of learning environments in recent years has increasingly emphasized the importance of interactive and experiential approaches that allow learners to engage with and interpret

cultural values. For instance, Muntean et al. [6] proposed an interactive tangible exhibit within a cultural heritage space that enables visitors to construct meaning through physical interaction, aiming to convey the values of the Musqueam Indigenous people. Rather than one-way explanations of value, they demonstrated that allowing visitors to interpret cultural meanings through their own actions and choices encourages deeper understanding and reflection.

Efforts to integrate cultural values into education have also become closely associated with the development of sustainable learning environments. Orphanidou et al. [22] argued that in today's increasingly digital society, utilizing cultural heritage as an educational resource can effectively foster learners' values and social development. Additionally, Buragohain et al. [23] introduced an initiative that uses metaverse technology to digitally reproduce and share cultural heritage in virtual environments. This approach proposes a new form of learning that allows learners to immerse themselves in culture and autonomously derive meaning from their experiences.

In this way, cultural learning that incorporates digital technology does not merely involve the one-way transmission of information. Rather, it represents a new direction in educational practice that supports learners in experiencing, thinking about, and engaging in meaning-making of cultural values.

This study also places importance on learners recognizing cultural values from their own perspectives and being able to share and reconstruct them with others. Accordingly, in the design of scenarios and tasks, efforts are made to ensure that learners have opportunities to encounter cultural values through experience and engage in meaning-making. In designing learning environments in digital contexts, it is essential to support not only the transmission of cultural values but also the processes through which learners reinterpret and engage with them.

2.1.3. Food culture learning

Food culture is a cultural practice that embodies regional history, lifestyles, and values, and it has increasingly attracted attention as a theme in cultural education. Ma [2] points out that eating behaviors and food culture in China are deeply influenced by social and historical contexts. Food culture involves not only nutrition and taste, but also serves as a "medium for conveying cultural meaning". These aspects of food often serve as opportunities for learners to reflect and empathize, making food culture an effective medium for promoting understanding of cultural values. Against this backdrop, educational practices related to food culture have taken a variety of forms.

First, the "exhibition and explanation" approach is often implemented in museums and cultural institutions. This includes exhibitions, seminars, and guided tours of the food production process [24]. In addition to exhibitions that systematically disseminate food culture rooted in the region or a specific field, there are also special exhibitions on food culture from various perspectives, including science and history [25]. These engaging activities foster public interest and encourage more people to participate in cultural learning.

Second, there is the "experiential" approach, which integrates actual food-related actions such as cooking and eating into the learning process. Kobayashi et al. [1] have developed a system that integrates interactive video elements with real meals. This allows users to immerse themselves in the natural beauty, changing seasons, and cultural connections that enrich the experience of Japanese cuisine while enjoying their meals. This type of design has been shown to enhance learners' cultural sensitivity and interest.

Third, food culture is also utilized in "participatory and multicultural" approaches, where it serves as a means of promoting intercultural education and social integration. For example, Kang et al. [26] conducted a study on immigrant women living in South Korea, emphasizing both the importance of understanding traditional food culture and the need for corresponding educational programs. Their findings

suggest that food culture is an effective educational resource for encouraging intercultural understanding and facilitating adaptation to everyday life. In modern society, where cross-cultural encounters have become routine, such re-recognition of culture and relationship-building serve as valuable opportunities for learning.

However, such learning activities often depend on face-to-face or on-site participation, making them susceptible to limitations in time and geography. This presents high barriers to participation, particularly for international learners or those in remote areas. To overcome these constraints and provide a more flexible and accessible means of cultural learning, the next section explores the use of VR in cultural education.

2.1.4. VR systems supporting cultural learning

In recent years, Virtual Reality (VR) technology has attracted increasing attention for its ability to virtually recreate cultural contexts and situate learners within them. By integrating visual, auditory, and spatial sensory input, VR provides immersive experiences that promote active learner engagement and have been shown to positively influence learning motivation and knowledge acquisition [11].

Li et al. [8] developed "DianTea", a VR game designed to enhance young people's understanding of traditional Chinese tea culture. This study deconstructed the process of cultural learning into multiple scenarios to improve the learners' cultural understanding and interest. This study presents a comparative experiment between their system and a browsing version of the game, and the findings demonstrate that the interactive system significantly enhances knowledge acquisition, engagement, and immersive experience.

Cheng et al. [27] investigated the use of virtual reality in language learning through the adaptation of "Crystallize", a 3D game designed for learning Japanese. This study focused on teaching cultural behaviors such as bowing in Japanese greetings. A comparison between VR and non-VR versions of the game indicated that VR significantly increased engagement.

These studies indicate that VR is highly effective in supporting cultural knowledge acquisition and enhancing learning motivation. In particular, embodied experiences within VR environments facilitate intuitive and affective understanding of cultural contexts. However, most of these studies are based on individual learning models and fall short in addressing deeper learning aspects such as the meaning-making and reflective interpretation of cultural values.

The goal of cultural learning goes beyond the acquisition of knowledge; it also involves enabling learners to recognize cultural values in a personally meaningful way. Mantovani et al. [11] argue that 3D virtual environments are particularly effective for supporting experiential meaning-making, and that the design of the learning environment plays a critical role in shaping the quality of learning. Boud et al. [15] also argue that reflection is essential for deep learning. It involves reexamining one's own perspectives in response to new experiences.

Therefore, even in VR-based cultural learning, it is important not only to present knowledge, but also to design learning mechanisms that support meaningful understanding of values and reflective thinking. This study aims to address this need by integrating VR environments with collaborative learning to explore a new type of learning environment that supports the understanding of cultural values.

2.2. Collaborative learning

2.2.1. Collaborative learning and reflection

Collaborative learning is a commonly used educational approach. The broadest definition of collaborative learning is a situation in which two or more people learn or attempt to learn something together [14]. Many studies have shown that this form of learning contributes to improved learning outcomes and increased engagement in learning [28].

Collaborative learning not only enables the sharing of knowledge but also promotes cognitive development through interaction among learners. In particular, the promotion of learner reflection is regarded

as an important cognitive benefit of collaborative learning [15]. Here, reflection refers to the thinking process in which learners review their own experiences and understanding and engage in meaning-making.

This reflection is mainly supported by verbal interaction within the collaborative learning process. For example, when learners engage in self-explanation, they become aware of incomplete aspects of their understanding and are able to revise their knowledge [29]. Furthermore, when learners externalize their thinking and express it to others, activities such as stating opinions, asking questions, interpreting, explaining, discussing, generating and testing ideas, and giving supplementary explanations are activated. These verbal and dialogic actions provide opportunities to visualize and share thoughts, thereby prompting reflection [16].

In this way, collaborative learning goes beyond mere information exchange and serves as an important framework for learners to reconstruct their understanding and advance toward deeper learning. This study also focuses on how collaborative interactions in VR environments influence the process of learner reflection and meaning-making of cultural values.

2.2.2. Gamification installations in museums supporting collaborative learning

As one example of collaborative learning support in museums, Nofal et al. [30] presented TouchTomb, a tangible gamification installation designed to enrich informal cultural learning experiences for young museum visitors. They combined tangible user interfaces with gamification methods that provide collaborative and engaging learning experiences to enhance learners' engagement and collaboration. A distinctive feature of its design is the incorporation of cultural knowledge as part of the game's reward structure, thereby naturally guiding learners toward information acquisition and enhancing their motivation.

This design philosophy aligns with the concept of "meaning-making" emphasized by Silverman [31], who argues that learning in museums is not merely about passive reception of information, but rather a process in which visitors interpret and reconstruct cultural meaning based on their own experiences and values. The inclusion of game elements that prompt learners to ask questions and make contextual judgments has proven effective in supporting this form of meaning-centered learning.

In addition, Hornecker et al. [32], through their analysis of interactive museum installations, have identified key insights into interaction design within public learning spaces. They highlight how the physical arrangement of exhibits, social configurations, and tangible interfaces influence learners' exploratory behaviors and the emergence of collaborative dialogue. These findings offer valuable implications for the design of gamified TUI-based exhibits and can also inform the development of collaborative learning systems in virtual environments.

This study adopts a similar design approach by embedding cultural information as a prerequisite or reward within learning tasks, aiming to encourage learners to actively explore cultural values and collaboratively construct meaning. As Nofal et al. pointed out, physical installations come with constraints related to cost, deployment flexibility, and durability. VR technology, however, offers a means to overcome these limitations and enables the sustainable operation of learning spaces.

Applying this gamification design concept to VR and constructing a learning experience that supports the exploration of cultural values and collaborative meaning-making is a critical direction for future development. This study aims to realize that potential by reconstructing a museum-like collaborative learning environment within virtual space.

2.2.3. VR systems supporting collaborative learning

In recent years, collaborative learning in VR environments has attracted increasing attention, especially in the domains of medicine and biology, where its effectiveness has been empirically validated. Drey et al. [33] compared two systems for pair-based learning in VR: a

symmetric system where both participants used VR, and an asymmetric system where one used a tablet. Their findings demonstrated that the symmetric system led to significantly higher levels of presence, immersion, and user experience. Based on these results, the researchers emphasized the importance of maintaining system symmetry in collaborative VR learning, such as by providing each user with their own avatar.

Collaborative VR learning has been widely adopted in scientific and technical disciplines, such as medical and biological education. De Back et al. [34] conducted a study using a Cave Automatic Virtual Environment (CAVE) for teaching neuroanatomy and found that collaborative VR learning yielded superior learning outcomes compared to traditional textbook-based methods. The effect was especially pronounced among learners with lower spatial recognition skills.

Schott et al. [35] developed a multi-user VR/AR system for liver anatomy education, enabling students and instructors to collaboratively explore complex anatomical structures using immersive visualizations grounded in clinical scenarios and patient data. Their study demonstrated that collaborative operations within VR significantly promoted interpersonal communication and social interaction among learners.

Chheang et al. [36] introduced CollaVRLap, a collaborative VR system designed for laparoscopic surgery training. This system integrates actual medical data and interactive surgical instruments in a shared immersive environment. Their results highlighted that collaboration between surgeons and assistants improved through enhanced communication and coordination in the VR setting.

These studies collectively underscore the effectiveness of collaborative VR learning in improving learning outcomes, learner engagement, and interpersonal communication. A systematic review by Radianti et al. [37] also identified presence, interaction quality, and social relatedness as key metrics used to evaluate collaborative VR learning. Dillenbourg [14] further emphasized the need to ensure interdependence and cognitive synchronization when designing collaborative learning systems. Slater [38] noted that presence in VR is strongly linked to learning outcomes, suggesting that a high degree of immersion and engagement in virtual environments directly influences the quality of collaborative learning.

However, it should be noted that most of these studies have been conducted in STEM-related fields such as medicine, biology, and engineering, where educational goals primarily focus on memorizing information and acquiring practical skills. In contrast, cultural learning requires not only knowledge acquisition and motivation but also learner reflection on cultural values and the process of engaging in meaning-making by connecting those values with their own experiences.

Roussou [39] demonstrated that learning through play in interactive virtual environments is effective for fostering value comprehension and concept formation in children. Through VR experiences involving ancient Greek mythology, history, and artifacts, children were able to intuitively understand cultural backgrounds and actively construct personal meaning. Pujol and Champion [40] evaluated the role of presence in cultural heritage VR learning and revealed its impact on learners' subjective understanding of cultural content. These studies provide important insights into the design and evaluation of collaborative VR learning within cultural contexts.

Considering these differences in characteristics, it is difficult to apply the designs and findings of collaborative VR learning developed in scientific fields directly to cultural education. This study explores how learners can gain a deep understanding of cultural values and engage in meaning-making from their own perspectives through collaborative learning in VR environments.

3. Design of food culture experience system

This study proposes a VR learning environment that supports scenario-based collaborative learning of food culture. Aiming to enable learners to interpret cultural values through collaborative experiences, this section describes the characteristics of Japanese food culture, the structure of the learning content, and the system design.

3.1. Characteristics of Japanese food culture and learning content

Japanese food culture is one of the most popular food cultures in the world. In 2013, Japanese food was recognized by UNESCO as an Intangible Cultural Heritage, which has four primary characteristics: (1) respect for diverse and fresh ingredients and their characteristics, (2) nutritional balance that supports a healthy diet, (3) expression of natural beauty and seasonal changes, (4) close relations to festival events. These features extend beyond cuisine to encompass cooking methods, tableware, and the handling of ingredients, giving Japanese food culture its multifaceted value [41].

Among these, the visual and cultural elements – such as the expression of seasonality and the design of tableware – are particularly distinctive aspects that characterize Japanese food culture and attract attention in cross-cultural comparisons. To enable learners to understand such cultural features through experiential learning, this study focuses on the theme of autumn and sets the learning content around three aspects: ingredients, tableware, and cuisine. Autumn was selected as the theme due to its abundance of natural scenery, seasonal ingredients, and traditional events, making it visually striking and well-suited for conveying seasonal charm in VR.

The selection of learning content was based on educational materials published by Japan's Ministry of Agriculture, Forestry and Fisheries [41–43], which aim to widely convey the appeal and values underlying Japanese food culture and contribute to cultural inheritance and international dissemination. Drawing on these resources, this study constructs three autumn-themed scenarios to help learners not only acquire knowledge but also understand the cultural values and meanings through immersive experience.

3.2. Learning environment composition

As shown in Fig. 1, this learning environment consists of three scenarios based on three themes related to Japanese food culture: “ingredients”, “tableware”, and “cuisine”. These themes reflect essential aspects of Japanese food culture, each deeply connected to specific cultural values—for example, ingredients reflect values such as respect for seasonality and harmony with nature; tableware embodies aesthetic sensitivity and the expression of seasonality; and cooking conveys values such as harmony in taste, color, and technique.

To support experiential understanding of these cultural values, this study adopts a scenario-based collaborative learning approach. Scenarios provide contexts through which learners can interpret meaning, and tasks encourage active participation, as well as the collaborative interpretation and sharing of cultural values. The goal is to facilitate meaning-making through experience, rather than treating cultural values as mere knowledge to be acquired.

Each scenario presents cultural values using different methods and at different timings. In Scenarios 1 and 2, related cultural information is provided during the task, without explicitly stating the intended cultural values. In contrast, Scenario 3 presents information about cultural values explicitly after the task to promote discussion and deepen interpretation. All tasks are grounded in familiar activities, such as collecting, observing, and cooking, and incorporate game elements to enhance learner motivation and immersion.

As shown in Fig. 2, the environment consists of outdoor autumnal nature scenes and indoor traditional Japanese architectural spaces. Scenario 1 takes place outdoors, allowing participants to feel the crisp autumn air and colorful foliage while exploring seasonal ingredients, thereby encountering the value of appreciating seasonal change. Scenarios 2 and 3 are held indoors in a gallery and kitchen setting, encouraging observation and hands-on experience with cultural values embedded in tableware and cooking. In particular, the beautiful recreation of autumn landscapes plays a key role in engaging learners' senses and fostering deeper cultural empathy and understanding through visual immersion.

3.3. Avatar design

3.3.1. Purpose of introducing avatars

To support collaborative learning in virtual spaces and promote communication between users, we introduced avatars as points of reference. By introducing features that provide social cues, such as voice and gestures, social interaction between users can be easier and more immersive [33].

Avatars visualize and embody the presence of others in the virtual space, providing learners with a sense of “co-presence” or sharing the same space. This sense of co-presence facilitates the formation of collaborative relationships and increases learners' engagement in dialogue and behavior. Furthermore, the combination of voice and gestures allows learners to express themselves not only verbally but also through body language and movement, lowering the barrier to participating in conversations.

3.3.2. Gestures and non-verbal communication

We anticipate that the gestures will fall into two categories: providing information and expressing emotions, for example, pointing to a specific object to provide information or giving a thumbs-up to encourage a partner.

These gestures promote joint attention and mutual understanding within VR spaces, complementing verbal communication by conveying emotions and judgments that may be difficult to express in words. The system supports the sharing of attention and interests among learners through nonverbal cues such as pointing and head orientation, thereby triggering dialogue and cognitive alignment. Although eye-tracking is not implemented, the direction of the head is reflected in the avatar's movements, enabling a rough estimation of the learner's gaze and producing a similar effect.

In fact, it has been reported that sharing gaze behaviors and gestures in VR and Mixed Reality environments enhances joint attention and collaborative behavior, thereby promoting the integration of understanding among learners [44,45]. The accumulation of such reciprocal interactions is said to increase the sense of social presence, making learners more acutely aware of the presence of others [46].

Furthermore, as social presence increases, learners become more aware of their own behavior through others' responses and gaze, which in turn promotes self-reflection. Soilis et al. [47] and Jiang et al. [48] have also pointed out that the presence of others and embodied experiences in VR can encourage learners to reflect on themselves and re-evaluate values, serving as a catalyst for critical reflection.

3.3.3. Technical implementation of avatars

To achieve the above requirements, we used full-body avatars to visualize their positions, movements, and hand gestures. Full-body avatars generate the highest level of co-presence and behavioral interdependence compared to other types of avatars [49].

We tracked the user's head with an HMD (head-mounted display) and their hands with two controllers, and we used IK (Inverse Kinematic) techniques to generate the user's upper body movements. Additionally, we used the sensor feature on the controller buttons to detect the position of the user's fingers. For instance, if the user places their index finger on the trigger button, it is recognized as the index finger being bent, thereby generating the hand gestures in virtual space. Since each finger aligns with a corresponding controller button of the avatar's finger in the controller's grip method, we did not provide instructions, as we believe that users can naturally grasp how to generate hand gestures by trying to touch the buttons (see Fig. 3).

On the other hand, to minimize the effects of simulator sickness, we decided to provide teleportation to allow users to move freely in the virtual space. Users could also use controllers to interact with the scene, such as grabbing objects.

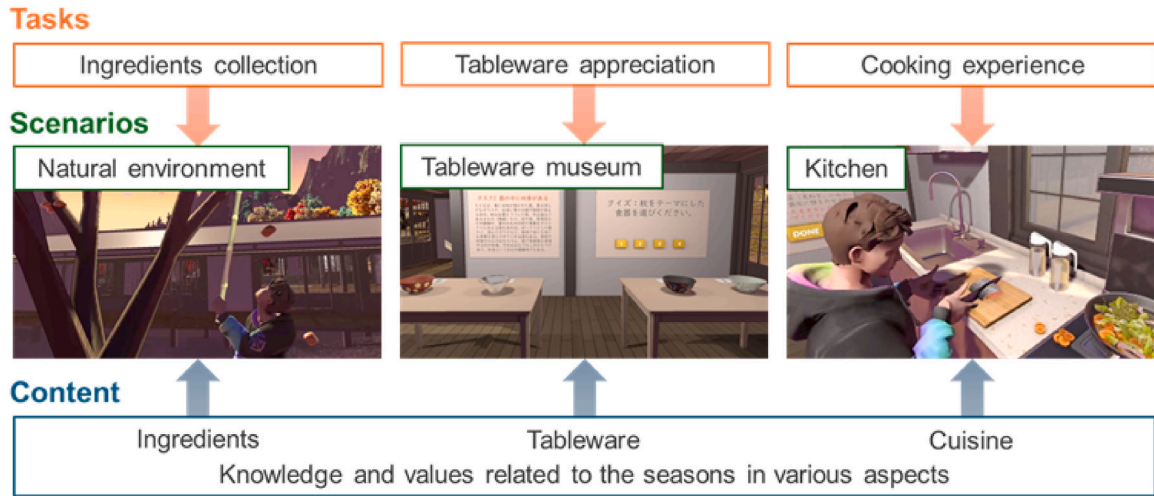


Fig. 1. The learning environment consists of three scenarios, each corresponding to three themes and tasks.



Fig. 2. (a) Top view of the entire virtual environment. (b) The natural landscape of the first scenario. (c), (d) Autumn ingredients such as salmon and persimmons are located in the natural environment.

3.3.4. Connection to research questions and link to evaluation

The introduction of avatars is directly connected to the verification of the research questions RQ1 and RQ2. In particular, the process by which learners share perspectives and insights through non-verbal gestures and head movements, and collaboratively construct the meaning of cultural values, is closely related to RQ1, which examines whether learning can progress without explicit instruction.

Furthermore, behavioral indicators such as the frequency and quality of verbal and non-verbal interactions, the timing of responses in dialogue, and the degree of synchrony serve as effective measures for capturing differences in learning outcomes and motivation between collaborative and individual learning. These indicators form the basis for comparative evaluation in RQ2.

3.4. Scenarios and task design

In this learning environment, learners sequentially experience the three scenarios. In each scenario, learners begin by checking the task

instructions displayed on a panel, then work together in pairs to complete the task. Upon completing a task, they proceed to the next scenario.

Each task is designed to naturally elicit collaboration between learners. The activities – such as exploration, observation, and manipulation – are structured to create opportunities for learners to pay attention to each other's actions and share insights and judgments, thereby promoting dialogue and information exchange.

Furthermore, the non-verbal expressions of avatars are closely linked to task progression. For example, during ingredient exploration, pointing gestures help draw attention; during the appreciation of tableware, responses are conveyed through nodding or head movements; and in the cooking scene, collaborative actions are synchronized with physical movements. These forms of interaction allow learners to infer their partner's intentions and perspectives without using words. The system also reflects head orientation, enabling the estimation of gaze direction and supporting the formation of joint attention.



Fig. 3. Full-body avatars visualize users' positions, movements, and hand gestures in the virtual space.

In this way, interactions that include non-verbal expressions serve to visualize learners' awareness and understanding, functioning as elements that facilitate meaning-making and reflection on cultural values.

3.4.1. Natural environment scenario

The theme of the first scenario is “ingredients”, and the cultural value we aim to convey is “Treasuring seasonal ingredients”. In this scenario, learners explore an outdoor virtual space that recreates the natural landscape of autumn in Japan, where seasonal ingredients are naturally embedded within the landscape. By combining 3D models of the ingredients with the natural setting, learners can visually recognize these ingredients in their natural state.

Some ingredients can be collected using nearby tools, introducing game-like interactions that enhance engagement and realism. When learners place the gathered ingredients into a basket, a panel displays relevant information such as nutritional benefits, historical usage, and related cultural traditions or dishes. For example, in the case of chestnuts, the system introduces production regions, antioxidant properties, and cultural backgrounds such as chestnut rice and chrysanthemum wine during the Chrysanthemum Festival.

Before the exploration begins, learners can refer to a map displayed in the environment and discuss with their partner how to divide areas and plan their search. This structure encourages collaborative decision-making, such as role-sharing and route planning. These dialogues not only increase the efficiency of the exploration but also offer opportunities to understand each other's perspectives and reasoning, thereby fostering mutual understanding and joint attention.

Learners work in pairs to explore and collect ingredients, engage in dialogue while reading the information panels, and share their prior knowledge and experiences. Especially during exploration, avatar gestures such as pointing or nodding enable learners to notify each other of discoveries and express reactions, helping to visualize shared awareness and mutual interest. These interactions turn the activity of finding and confirming ingredients into a collaborative process of meaning-making rather than a mere task.

Moreover, the information about ingredients offers not only new knowledge but also connections to learners' own lives—such as ingredients they are familiar with, how those are used at home, or how practices vary by region. This encourages culturally grounded dialogue to emerge naturally, allowing learners to reinterpret values in relation to their own backgrounds.

Through this design, learners are expected to engage in reflective understanding of the cultural value of “seasonal ingredients” through

embodied experience, peer dialogue, and reflective thinking. Additionally, the natural environment recreated in this scenario functions not just as visual decoration, but also as psychological space that supports calm reflection. Jiang et al. [48] have pointed out that immersion in VR can sometimes hinder reflection, emphasizing the importance of designing experiences that allow users to step back and reinterpret what they have experienced. This scenario, by providing a gently stimulating natural setting, encourages learners to quietly reflect on their actions and cultural background, functioning as a reflective moment that supports experiential meaning-making (see Fig. 4).

3.4.2. Tableware museum scenario

The theme of the second scenario is “tableware”, and the cultural value we aim to convey is “The aesthetic sense of visually expressing the beauty of seasons”. In the exhibition room, a variety of tableware items themed around the four seasons – spring, summer, autumn, and winter – are displayed. Learners can freely observe and appreciate these items while gaining an understanding of how seasonality is represented and how design elements are applied in Japanese food culture.

Additionally, a quiz is implemented in which learners match each piece of tableware with its seasonal theme. Learners collaboratively examine and discuss visual features as they consider their answers. Information panels provide background on the tableware, including its origin, the meanings of its patterns, and the intentions behind its use of color. With this information, learners are encouraged to share and negotiate their observations and interpretations with one another.

Because impressions and interpretations of tableware may vary from person to person, this task naturally fosters collaborative interaction. For example, one learner may comment, “This bowl feels like autumn”, and the other may respond, “That might be because of the maple leaf pattern”. In this way, the exchange of meaning-making based on visual elements occurs naturally. Moreover, non-verbal reactions by avatars – such as tilting the head or nodding in front of a display – serve as cues that indicate agreement, hesitation, or levels of understanding, thereby complementing spoken dialogue and enriching the quality of interaction.

Furthermore, gestures such as pointing and the alignment of head direction help learners share their attention toward the same object, facilitating joint attention and the subsequent alignment of perception. Through this process, understanding of cultural values is not treated as mere knowledge acquisition, but rather as a collaborative process of meaning construction through dialogue and observation (see Fig. 5).

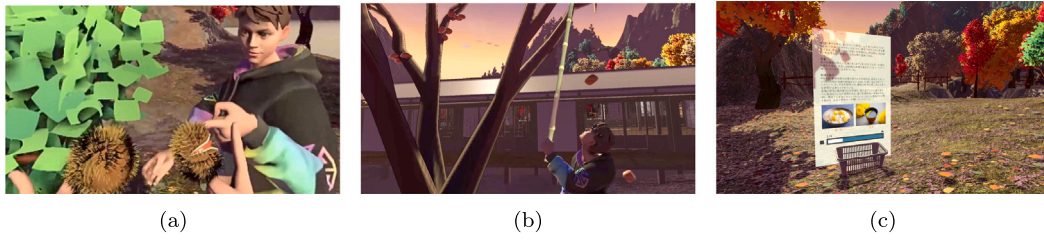


Fig. 4. The First Scenario: (a) Avatars are sharing information about ingredients. (b) Avatar is using a tool to collect ingredients. (c) After containing ingredients, relevant information will be displayed on a panel.

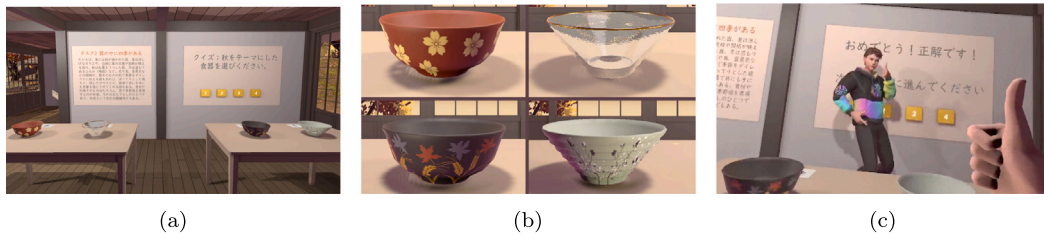


Fig. 5. The Second Scenario: (a) A variety of tableware is displayed in an exhibition room. (b) These pieces of tableware are designed with different seasons as their themes. (c) Avatars need to solve a quiz matching the seasonal theme with the tableware.

3.4.3. Kitchen scenario

The theme of the third scenario is cuisine, and the cultural value to be conveyed is the “harmonious balance among taste, colors, and cooking techniques”. In the kitchen scene, kitchenware, ingredients, and recipe panels based on traditional Japanese dishes are placed, allowing learners to cook together in pairs while experiencing the process of Japanese cooking and the cultural values behind it.

Cooking operations are simple and consist of basic actions such as slicing vegetables with a knife or adding ingredients to a pot. During the cooking process, information is displayed about the types and purposes of kitchen tools, how heat is applied, and the care taken in each step. Learners can thus learn about cultural values such as the “delicate attention to process” and the “culture of tool use” in Japanese cuisine. When the dish is completed, it appears on the dining table along with a score, and a panel displays information explaining the cultural value of “harmonious balance among taste, appearance, and techniques”.

In this scenario, learners cook through dialogue and cooperation. They are expected to ask each other questions such as “What should we do next?” or “Which tool should we use?” However, cooking is performed one at a time rather than simultaneously, and there are no predefined roles. Because of this, dialogue and consensus during the task become the main triggers for collaboration.

Learners can also communicate through non-verbal cues – such as gestures and reactions – during the cooking process. For instance, nodding while watching the partner’s actions or offering a hand gesture can help communicate intention and understanding without explicit instructions. These physical interactions help foster a sense of empathy and cooperation between learners, which can contribute to a deeper understanding of the cultural value.

Finally, by reading the cultural information presented after cooking and reflecting on their shared experience, learners are given the opportunity to reflect on the value of “harmony among taste, color, and technique” based on their embodied experience. Through this integration of physical activity, peer dialogue, and reflective prompts, this scenario is designed to support a deeper understanding of cultural values (see Fig. 6).

3.5. System implementation details

The VR learning environment was developed using Unity 2022.3.8 LTS and deployed on Windows PCs using Meta Quest 2 head-mounted displays and Oculus Touch controllers as input devices.

Networking and real-time communication were supported using Photon Fusion to enable multi-user collaboration. To support voice communication between paired users, Photon Voice was used to provide low-latency audio streaming.

4. User study

4.1. Study design

We conducted a comparative experiment to answer RQ1 and RQ2. The participants were randomly divided into two groups. One group underwent individual experiences, while the other participated in paired experiences.

We measured learning outcomes as follows. A questionnaire with pre- and post-experience modules was designed to assess participants’ understanding of related knowledge and values. After participants completed the questionnaire before and after the experience, we evaluated the learning outcome by comparing the results of these two modules. To confirm the influence on the participants, we asked them to answer a follow-up questionnaire one week after the experience to evaluate their awareness of learning and thinking about food culture in their daily lives.

To further understand and analyze the collaboration between participants during collaborative learning, and to obtain additional evidence for the two RQs, we also recorded videos of the paired group experiences and documented and categorized their collaboration (see Table 1).

To measure participants’ motivation in their learning experiences, we used the Intrinsic Motivation Inventory (IMI) [17]. The IMI has demonstrated strong validity and reliability [18]. The scales used in the questionnaire comprise Interest/Enjoyment and Value/Usefulness.

4.2. Procedure

After recruiting participants, we randomly assigned them to different groups. Participants initially answered a pre-experience questionnaire, after which they watched an introduction video to learn how to operate the VR system. Participants then entered the system individually or in pairs. After the experience, the participants answered a post-experience questionnaire. One week after the experience, participants answered a follow-up questionnaire.



Fig. 6. The Third Scenario: (a) Kitchenware and ingredients are placed in a kitchen scene. (b) Avatars can cook according to a recipe with simple operations. (c) After cooking, the finished dish and an introduction to the cuisine will be displayed on the table.

Table 1

Questions included in the learning effectiveness questionnaire.

Before experiencing the system	What do you think of Japanese food? Please describe Japanese food as much as you can.
After experiencing the system	What did you learn from this experience? What kind of values do you think Japanese food has?
One week after the experience	Did you become more conscious and change something in your daily diet? Have you ever been aware of or acted on food-related events, seasons, nature, tableware, ingredients, etc., in your daily life?

4.3. Participants

Participants were recruited through advertising posters targeting students within the university. To be eligible, participants were required to be over 18 years old, fluent in Japanese, and available during the experimental period.

A total of 16 participants took part in the study. Eight were assigned to the individual condition, and eight to the paired condition. Among them, 12 reported prior experience with VR head-mounted displays. Although all participants received a tutorial before the session, differences in VR proficiency may have influenced how easily they interacted or how well they focused on the tasks within the virtual environment.

It is also important to note that participants shared relatively homogeneous characteristics in terms of age, academic background, and cultural background. These factors may have introduced certain biases in how cultural values were perceived and interpreted, which should be considered when generalizing the findings.

Although the sample size ($N = 16$) is relatively small, we complemented the quantitative analysis based on questionnaire responses with a structured behavioral coding and analysis of participants' collaborative interactions during the paired sessions. This approach provided deeper insight into how collaboration contributed to the spontaneous interpretation and understanding of cultural values (RQ1), as well as how specific collaborative behaviors influenced learning outcomes and motivation (RQ2). It also offered additional qualitative perspectives beyond what could be captured through questionnaires alone. Considering the combination of quantitative analysis and in-depth behavioral observation, we believe the sample size of 16 participants is sufficient to address the research questions and derive meaningful insights.

5. Results

5.1. Learning outcome

5.1.1. Before experience

For comparison, in the pre-experience questionnaire, we asked: "What do you think of Japanese food?" and "Please describe Japanese food as much as you can".

In the answers to the first question, three participants mentioned "healthy" and "balanced nutrition". Another three responses referenced the global acclaim and cultural significance of Japanese cuisine. Two

participants mentioned the diversity of Japanese cuisine, including its raw food culture with dishes like sashimi. Additionally, two participants highlighted the cuisine's delicate and subtle flavors, while another felt the seasoning of Japanese food was too light. In the second question to confirm knowledge, their answers included brief statements about related traditions, etiquette, and uniqueness.

As the responses reflect, all participants' understanding of Japanese food culture centers primarily on these more widely recognized aspects.

5.1.2. After experience

In the post-experience questionnaire, we asked, "What did you learn from this experience?" and "What values do you think Japanese food has?"

(1) Gained Cultural Knowledge

In the questionnaire, we asked, "What did you learn from this experience?" To assess the cultural knowledge acquired by participants, answers were evaluated based on content relevance and level of detail. Based on content specificity, responses were classified into three levels: irrelevant, generalized, and detailed. An irrelevant answer refers to a response that is unrelated to Japanese food culture or where the connection to the learning content cannot be determined. Examples include statements such as "Japanese food is delicious" or "I like eating rice". A generalized answer is one that is related to Japanese food culture but remains vague, providing limited information without specifying the acquired knowledge. These responses typically contain only a broad description of a single aspect without concrete facts or supporting details, such as "Food culture related to the seasons" or "The history of Japanese cuisine". A detailed answer, on the other hand, is a response that is highly relevant to the learning topics, such as ingredients, tableware, and cuisine, and includes at least two specific pieces of information. Examples of detailed answers include statements such as "Japanese persimmon is also called 'KAKI' overseas and is rich in beta-carotene, which is beneficial for health" and "Japanese cuisine incorporates the Yin-Yang and Five Elements philosophy, stimulating all five human senses".

All participants in the paired experience group provided detailed answers related to the learning content. Six participants mentioned ingredients. Three of them mentioned the importance of seasonality, noting how it influences the taste and selection of ingredients, such as salmon, persimmons, and chestnuts in autumn. Three of them mentioned the geographical origin and history of use. Four participants

mentioned the seasonally themed tableware in Japanese food culture, emphasizing how the designs of plates and bowls convey a sense of seasonality. All eight participants mentioned Japanese cuisine and cooking methods. Four of them mentioned the preparation and origin of the traditional dish “Chan Chan Yaki”. Five of them mentioned various types and uses of Japanese kitchen knives. One of them expressed interest in seasonal dishes such as “Kaki Namasu”.

In contrast, only three participants in the individual experience group provided detailed answers related to the learning content. Two participants gave generalized answers, such as “Food culture related to the seasons” or “The history of Japanese cuisine”. The other participants provided irrelevant answers.

(2) Recognition of Cultural Value

In response to the question, “What kind of values do you think Japanese food has?” Five participants in the paired experience group mentioned “treasuring seasonal ingredients”. Three participants mentioned “visually expressing the beauty of seasons”. In addition, two participants mentioned the value of careful consideration of even the smallest details, and one participant mentioned the value of coexistence with nature.

On the other hand, three participants in the individual experience group mentioned valuing seasons and ingredients, while the others provided answers unrelated to food culture.

5.1.3. Follow-up survey

To investigate the impact after the experience, we asked them to answer a follow-up questionnaire one week after the experience. The questions include: “Did you become more conscious and change something in your daily diet?” “Have you ever been aware of or acted on food-related events, seasons, nature, tableware, ingredients, etc. in your daily life?”.

Responses from participants in the paired experience group reported notable behavioral changes and increased awareness. Five of them noted that they became more conscious of choosing seasonal and fresh ingredients and checking the origins of their food more carefully. Two of them noted that they paid attention to cooking techniques like those seen on cooking shows and showed increased interest in researching recipes with seasonal produce.

In contrast, two participants from the individual experience group mentioned buying seasonal vegetables and seasonal-themed tableware. The other participants responded that they had no particular changes or awareness.

5.2. Motivation

A one-tailed Wilcoxon rank sum test revealed a significant difference ($W = 13, p = 0.023$) between the two groups, indicating that the motivation levels in paired experiences are significantly higher than in individual experiences.

5.3. Collaboration records

To better understand how collaborative learning influences the learning process and to provide additional evidence for the two research questions, this study recorded the activities of participants in the paired condition through video and documented collaboration based on observation.

5.3.1. Classification of collaborative behavior

In this study, collaborative behavior was documented and categorized from two aspects: verbal expressions and gestures. Verbal expressions were classified into five types – *knowledge sharing*, *expressing feelings*, *guiding*, *encouraging*, and *solving together* – while gestures were categorized into two types: *informational gestures* and *encouraging gestures*. These classifications were designed based on previous research on constructive dialogue and social talk in collaborative learning [50,51],

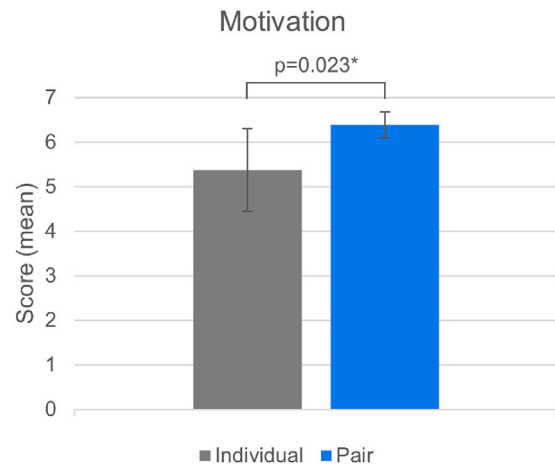


Fig. 7. We found a significant difference in motivation between the two groups.

as well as findings on the role of bodily expression in multimodal interaction [45,52], and were refined through preliminary observations conducted in this study.

Verbal Expression Types (5 categories):

- *Knowledge sharing*: utterances in which learners share their own knowledge, experiences, or opinions.
- *Expressing feelings*: utterances that convey direct impressions of a target, e.g., “I think salmon is really delicious”.
- *Guiding*: unilateral instructions to the partner regarding how to proceed with the task or operate the system.
- *Encouraging*: positive utterances such as praise or support toward a partner’s actions.
- *Solving together*: utterances in which learners discuss how to proceed with the task together.

In terms of verbal expressions:

Knowledge sharing refers to utterances in which learners talk about cultural objects based on their own experiences and knowledge, often serving as opportunities for meaning-making and reinterpretation.

Expressing feelings involves utterances that express impressions or memories about the object, contributing to the internalization of cultural values through emotional engagement.

Guiding is characterized by one-sided instructions about task procedures or operations and serves as an indicator of the task structure and the relationship between learners.

Encouraging refers to positive utterances that express praise or support for the partner, which contributes to maintaining a collaborative atmosphere and motivating learners.

Solving together refers to interactions where learners collaboratively consider cultural meanings and how to proceed with the task. It is regarded as a core activity in constructive learning.

Gesture Types (2 categories):

- *Informational gestures*: actions that communicate information, such as pointing or drawing attention to an object.
- *Encouraging gestures*: physical expressions such as a thumbs-up used to support or encourage the partner.

The classification of gestures in this study was constructed based on McNeill’s gesture theory [52]. In particular, *informational gestures* correspond to *deictic gestures*, such as pointing or guiding the partner’s gaze, which help direct learners’ attention to specific objects and support the formation of joint attention. Similarly, *encouraging gestures* correspond to *emblematic gestures*, such as giving a thumbs-up or clapping, which function as expressions of social engagement and empathy, softening

Table 2

Average frequency of each collaboration per person by scenario.

Type	Category	Scenario 1	Scenario 2	Scenario 3	Total
Speech	Knowledge Sharing	4	0	0.25	4.25
	Expressing Feelings	2.375	0.5	0.125	3
	Guiding	2.375	0	1.5	3.875
	Encouraging	0.25	0.25	0.125	0.625
	Solving Together	3.875	1.625	0.875	6.375
Gesture	Informational	1	0.5	0.25	1.75
	Encouraging	0.5	0.125	0.25	0.875

the atmosphere of dialogue and promoting sustained interaction. In our observation, we also confirmed multiple scenes where these gestures functioned as triggers for verbal interaction and meaning-making of cultural values—for example, when a participant said, “This feels like spring”, accompanied by a pointing gesture.

5.3.2. Quantitative analysis of types and frequencies of collaborative behavior

We calculated the average number of each type of collaborative behavior per participant in each scenario, as shown in Table 2. Coding was conducted consistently throughout the analysis. Classification criteria were based on prior studies [50,52], and refined through preliminary observations. Only clearly observed behaviors were recorded, with attention to ensuring consistency and reproducibility.

Scenario 1: Ingredient collection. The most frequent type of collaborative behavior observed was *solving together*. Participants worked together to locate ingredients and discuss how to collect them. In contrast, *knowledge sharing* and *expressing feelings* appeared frequently in the phase after collection, when participants were reading cultural information. In particular, *knowledge sharing* was prominent as learners exchanged newly acquired and prior knowledge. They discussed which seasonal ingredients they found especially tasty, how to cook them, and which seasonal dishes they regularly prepared or ate at home.

Expressing feelings was also clearly observed, as participants shared their preferences or memories related to seasonal ingredients or seasonal meals. These interactions were particularly active during the phase of reading cultural information after the exploratory experience. This sequence of experience, information presentation, and dialogue appeared to encourage learners to reinterpret cultural values and construct meaning.

Scenario 2: Tableware appreciation. In this scenario, the dominant behavior remained *solving together*. Learners discussed which seasonal motifs the tableware patterns represented and how they conveyed seasonality. For example, some participants focused on bowls with cherry blossom designs and discussed their symbolic representation of spring in Japanese culture.

Regarding *expressing feelings*, learners shared sensory impressions such as “this piece evokes a feeling of coolness”, highlighting their emotional reactions to specific tableware.

Scenario 3: Cooking experience. The dominant collaborative behavior in this scenario was *guiding*, particularly as participants prompted their partners on the next steps in the cooking process. However, during the phase of reading cultural information after completing the cooking task, there was little to no interaction or dialogue between participants.

This suggests that when the task structure is centered on procedural operation, utterances tend to be biased toward *guiding*, with fewer opportunities for dialogue involving *knowledge sharing* or cultural reinterpretation. These findings indicate that fostering reflective meaning-making requires a learning design that offers a certain degree of psychological space and moments to contemplate questions.

5.3.3. Overall observations

Encouraging behavior was observed mainly at points of task completion or when a partner proposed a solution. These included both verbal praise and non-verbal gestures such as a thumbs-up.

Informational gestures, such as pointing, nodding, or shaking the head, were used to draw the partner’s attention or express agreement/disagreement. Coordinated gaze and responses to pointing were frequently observed, indicating that learners shared attention and interest through head movement and other non-verbal cues. These actions often initiated dialogue and facilitated cognitive synchronization, suggesting that the system supported the formation of joint attention.

Clear behavioral differences were found across scenarios. Scenario 1 featured a high frequency of *solving together*, *knowledge sharing*, and *expressing feelings*, especially in the phase of reading cultural information. In Scenario 2, dialogue also centered on *solving together*, with a focus on visual and sensory characteristics of the tableware. In Scenario 3, however, utterances were mainly categorized as *guiding*, with little discussion of cultural information after the task.

These results indicate that task structure strongly influences the quality of collaborative behavior and that the timing and format of information presentation play a key role in activating learner dialogue. In scenarios where information followed directly after exploration or observation, learners naturally engaged in *knowledge sharing* and *expressing feelings*, which fostered reinterpretation of cultural values. Moreover, non-verbal behaviors such as avatar-mediated pointing and synchronized gaze contributed to the formation of joint attention, facilitating the timing of dialogue. These findings suggest that shared attention and interest among learners enhance the occurrence and quality of collaborative dialogue.

5.3.4. Reflections in verbal expressions

To further examine the internal processes of learners during collaborative VR experiences, we analyzed verbal expressions based on the three phases of reflection proposed by Boud et al. [15]: returning to experience, attending to feelings, and re-evaluating the experience. These categories were used to code learners’ spontaneous reflections that emerged during interaction.

Table 3 summarizes the reflection phases, coding criteria, and representative verbal expressions.

6. Discussion

This section discusses the learning support effects of the proposed system based on the results of the user study. Section 6.1 addresses RQ1 by analyzing how learners interpreted cultural values, drawing on observation results and a reflection model. Section 6.2 focuses on RQ2 and analyzes the effects of collaborative learning using questionnaire and observation data. Section 6.3 explores the role of gestures and avatars, while Section 6.4 summarizes the educational advantages of VR environments. Based on these findings, Section 6.5 presents design recommendations. Section 6.6 outlines the limitations of the study and future directions. Finally, Section 6.7 examines the system’s transferability to other cultures.

6.1. Interpret cultural values through collaboration

This section addresses RQ1: “How can scenario-based VR experiences facilitate learners’ understanding of cultural values in Japanese food culture through collaborative learning, without direct instruction?” To examine how the presentation format of cultural values influences learners’ meaning-making of their experiences, and how dialogue with others supports reinterpretation and internalization, this study compared three different scenarios. In Scenarios 1 and 2, cultural values were not explicitly presented, allowing learners to interpret them through their own experiences. In contrast, Scenario 3 explicitly presented cultural values using information panels.

Table 3
Reflection coding categories and observed example verbal expressions.

Reflection phase	Coding criteria	Representative verbal expressions
Returning to experience	Utterances where learners link the VR content or situation with their own life experience or memory.	“We often eat this at home.” “This bowl looks like the one at my grandmother’s house.”
Attending to feelings	Emotional or impressionistic responses, such as expressions of beauty, enjoyment, or familiarity.	“It’s really beautiful.” “This feels refreshing.” “This was seriously delicious.”
Re-evaluating the experience	Utterances where learners reconstructed their understanding or meaning through dialogue, often referring to cultural interpretation.	“Sensing the seasons through tableware feels very Japanese.” “Maybe this pattern is for spring.” “Seasonal ingredients really are full of good stuff.”

Scenarios 1 and 2 were designed in alignment with the reflection model by Boud et al. [15], which outlines three phases of learning reflection: “returning to experience”, “attending to feelings”, and “re-evaluating the experience”. For example, after learners explored ingredients, cultural information panels were presented. This allowed them to naturally look back on their experiences and prompted spontaneous dialogue linked to emotional impressions. In the case of tableware, the design encouraged learners to freely express their impressions and associations with the patterns, which helped foster sensory understanding and reinterpretation of cultural values. As analyzed in Section 5.3.4, learners’ verbal expressions aligned with these three reflection phases, supporting the applicability of Boud’s model to collaborative meaning-making in this context.

This design enabled learners to discover new perspectives and helped them notice cultural meanings that may not be easily recognized individually. In Scenario 1, for instance, one learner said, “My family often makes chestnut rice in the fall”, to which another responded, “Me too, it’s really delicious”. These conversations helped link personal memories and feelings with seasonal ingredients. In Scenario 2, one learner pointed to a bowl and said, “These cherry blossom patterns remind me of vibrant spring”, and the partner replied, “That’s true. In Japanese culture, cherry blossoms often symbolize spring”. These dialogues helped learners collaboratively construct an understanding of cultural aesthetics by connecting seasonal imagery with visual design. These interactions were supported not only by verbal cues but also by shared visual attention and non-verbal expressions, contributing to joint attention and collaborative meaning-making.

Questionnaire responses further supported this interpretation. Participants in the paired condition were able to express cultural values in their own words, such as “expressing seasonal beauty through tableware” and “treasuring seasonal ingredients”, indicating that they had reconstructed cultural values through their experiences and dialogue.

In contrast, few participants in the individual learning condition demonstrated recognition of these cultural values. This suggests that dialogue based on alternative perspectives and empathic understanding is essential for the reflective understanding of cultural values.

In Scenario 3, although active discussions were observed regarding the cooking procedures, very few verbal expressions related to cultural values emerged. One possible reason is that the values presented were too abstract and had weak relevance to the learners’ experience. In addition, the completion of the cooking task may have created a sense of achievement, which discouraged further open-ended dialogue or reflective verbal expressions. Furthermore, because the task structure required learners to focus on operations, most verbal expressions tended to be categorized as *guiding*, limiting opportunities for *knowledge sharing* or cultural reinterpretation.

Such differences appear to be rooted in the presentation method of cultural values. Scenarios 1 and 2 were designed to promote a learning process in which learners autonomously discover and construct meaning through experience, dialogue, and reinterpretation. In contrast, Scenario 3 presented cultural values only through abstract text, providing limited opportunity for learners to reinterpret those values in relation to their own experiences. This indicates that the phases of “attending to feelings” and “re-evaluating the experience” were not sufficiently activated in Scenario 3.

Quantitative analysis further supports these trends. In Scenarios 1 and 2, there were notably frequent verbal expressions categorized as *solving together*, *knowledge sharing*, and *expressing feelings*. In Scenario 1 in particular, active dialogue occurred after learners read cultural information following the exploration phase. This suggests that the sequence of experience, followed by information presentation, and then dialogue, effectively supported the reconstruction of cultural values. In contrast, Scenario 3 showed a higher proportion of *guiding* expressions, while instances of *knowledge sharing* and *expressing feelings* were much less common. Even after task completion, there was little dialogue related to cultural values, indicating that the task structure and information delivery timing suppressed reflective cultural discussion.

Based on these observations and analyses, we conclude the following regarding RQ1: Even without direct instruction, collaborative processes supported by scenario and task designs that include exploration, dialogue, and reflection can facilitate learners’ spontaneous reinterpretation and internalization of cultural values.

6.2. Effects of collaborative learning on learning outcomes and motivation

To answer RQ2: “What is the impact of collaborative learning, compared to individual learning, on learning outcomes and motivation in VR cultural learning environments?”, we analyzed the results from questionnaires and observations.

Previous studies have mainly focused on individual learning, emphasizing knowledge acquisition and engagement in VR cultural learning cases [8,27]. In contrast, the educational objective of this study includes not only the understanding of cultural values but also the cultivation of awareness toward cultural elements in daily life. Therefore, we also examined how collaborative learning influences these aspects.

Responses from the post-experience questionnaire suggest that participants in the paired condition more frequently reflected on cultural values during or after Scenarios 1 and 2, whereas participants in the individual condition often gave unrelated or superficial answers. In terms of both knowledge acquisition and post-experience impact, the paired condition showed superior learning outcomes and behavioral changes.

These results, together with the statistical analysis of motivation shown in Fig. 7, demonstrate that compared to individual learning, collaborative learning had significant effects on acquiring cultural knowledge, recognizing cultural values, increasing learning motivation, and promoting cultural awareness in daily life.

This difference is believed to stem not just from the amount of learning or accuracy of responses but from the ability of learners to reconstruct their understanding and interest by sharing perspectives and experiences with others during the learning process. In the subscales of the Intrinsic Motivation Inventory (IMI), “Interest/Enjoyment” and “Value/Usefulness” were rated highly, indicating that learners found meaning in the learning activities and that meaningful learning processes were taking place.

Observational records also confirmed many scenes in which participants explained things to their partners to organize their own understanding, or were inspired by each other’s reactions to develop the conversation. These kinds of social interactions are considered to have contributed to reflective thinking and deepening of understanding.

To further explore the perceived value of collaboration, two additional questions were included in the paired condition questionnaire. These questions asked whether participants preferred experiencing the system with others and what aspects of the experience they found particularly appealing or meaningful. All participants in the paired group responded that they would rather experience the system with others, such as family or friends, than alone. Additionally, seven participants wrote comments such as “collaborating made the experience more enjoyable” and “my partner’s comments enriched my understanding of various aspects of Japanese food culture”. These qualitative results further support the positive effects of collaboration on both knowledge acquisition and motivation.

Based on these results, we conclude that in response to RQ2, collaborative learning not only enhances learning outcomes and motivation but also contributes to deeper learning by reconstructing cognition through interaction with others and rediscovering the value of learning.

6.3. Using avatar to support communication

This system adopts a design that utilizes social cues such as avatar gestures and head orientation, and its effectiveness in supporting natural communication during collaborative learning in a VR environment was confirmed.

Observations showed that many participants in the paired condition spontaneously used gestures via controller input, without prior instruction. They effectively engaged in mutual interactions by identifying the partner’s focus of attention and conveying intentions through movements. Such nonverbal expressions helped establish shared understanding, even in situations where verbal speech was limited, thereby facilitating the smooth progress of learning activities.

The observed gestures were broadly classified into two categories: *informational gestures* and *encouraging gestures*. Examples of *informational gestures* included waving to call attention or pointing to patterns on a tableware to guide the partner’s focus. *encouraging gestures* included raising a hand in joy after completing a task or giving a thumbs-up to encourage the partner. These gesture expressions correspond to the functional classification proposed by McNeill [52]. Specifically, *informational gestures* are interpreted as *deictic gestures*, while *encouraging gestures* are considered as *emblematic gestures*. The former contributed to the formation of joint attention and visual synchrony, while the latter helped sustain collaboration and promote positive dialogue. For instance, in a scene where a participant pointed to a pattern on a piece of tableware and asked, “Isn’t this a cherry blossom for spring?”, the nonverbal act of pointing triggered the verbal utterance and helped guide the flow of conversation. Questionnaire responses also revealed that many participants felt encouraged by nonverbal gestures even in the absence of verbal praise, suggesting that these nonverbal expressions contributed to emotional reinforcement, thereby sustaining learners’ motivation.

The visualization of gestures and head orientation functioned not merely as auxiliary tools, but as essential components for learners to share attention, emotions, and thoughts. In particular, the ability to estimate the gaze target from head orientation facilitated the establishment of joint attention, enabling smoother topic development and timing coordination. As a result, it became easier to initiate speech and switch topics, improving the quality of collaborative work.

Furthermore, observation records showed that nonverbal interactions such as pointing at an object, gaze alignment, and nodding contributed to initiating speech and adjusting timing. Several instances were confirmed where speech began following synchronized gaze or where gestures were coordinated with verbal utterances. These findings suggest that the avatar’s bodily expressions naturally formed joint attention and served as a foundation for collaborative learning.

These observational results are consistent with prior research findings [44,46], which indicate that presence and co-presence in VR contribute to learners’ social engagement and motivation.

6.4. Advantages of VR in cultural learning

In this study, we constructed a scenario-based VR learning environment focused on Japanese food culture, focusing on three aspects: ingredients, tableware, and cuisine. Based on observations and questionnaire results, this section discusses the educational benefits of VR in cultural learning.

6.4.1. Feeling culture through experience

A major feature of VR is that it allows learners to engage with cultural content through embodied and immersive experiences. For example, in Scenario 1, participants engaged in an exploratory activity of walking through an autumn-themed natural setting while searching for seasonal ingredients. During the experience, they engaged in conversations like “Persimmon is called ‘kaki’ even overseas, right?” or “We make chestnut rice at home”, connecting their personal memories and everyday experiences to cultural knowledge. This kind of meaning-making through experience is difficult to achieve in conventional classroom or online educational settings.

Furthermore, free-form responses also included many comments such as “The exploration was fun” and “I was surprised by the beauty of nature and the tableware”, indicating that sensory-rich and immersive experiences through VR enhanced learner engagement and intrinsic motivation.

6.4.2. Experiencing multiple cultural elements in an integrated way

VR enables learners to experience cultural elements that are typically separated by time or space in the real world as one continuous and integrated experience. For example, elements like autumn landscapes, tableware displays, and traditional Japanese cooking were seamlessly integrated into a unified spatial environment, allowing learners to understand the interconnections between them. This integration is a unique advantage of the flexible learning environment that VR can provide.

Even during collaborative learning activities, VR demonstrated its effectiveness. Participants shared visual attention and restructured meaning together through interactions such as pointing to a tableware pattern and saying, “This looks like cherry blossoms—feels like spring, doesn’t it?” or asking, “Which season do you think this pattern represents?” These instances of joint attention contributed to a heightened sense of social presence as described by Biocca et al. [46], fostering engagement and learner connection.

6.4.3. Comparison with other collaborative learning methods

Collaborative learning itself can be conducted without VR—in classroom-based group work or online conferencing environments [14]. However, VR offers the advantage of visualizing the learner's attention and intentions through avatar-based nonverbal expressions, such as pointing, nodding, and head orientation. This reduces misunderstandings between learners and facilitates smoother, more intuitive dialogue.

For instance, when a learner pointed at a specific ingredient or tableware while saying, “This feels like spring”, the synchronization between gaze and gestures served as a trigger for dialogue and supported the construction of cultural meaning. Such real-time, spatially co-present experiences are educationally powerful in supporting the understanding of cultural values and reflective learning [45].

6.4.4. Differences from digital learning materials and unique features of VR

In classroom or online lectures, cultural values are typically conveyed through textbooks and video materials, using verbal and visual means [53]. In contrast, in a VR environment, learners engage with cultural objects through multiple senses, including visual, auditory, and spatial modalities, achieving an immersive “entry into cultural situations” that transcends mere information delivery.

This type of experiential immersion becomes a foundation for the internalization and reflective understanding of cultural values [45,54]. Nonverbal interaction through avatars, such as joint attention and mutual understanding, also facilitated natural dialogue and served as triggers for reflection.

In this way, cultural learning supported by VR provides learners with insights and experiences that are difficult to obtain through other methods, significantly enhancing the quality of learning.

These findings demonstrate that VR provides an effective environment for cultural learning by fostering understanding through experience, promoting collaborative dialogue, enhancing learning motivation, and supporting deep internalization of meaning. It provides educational value by enabling in-depth understanding and personal meaning-making that are difficult to achieve through other approaches.

6.5. Design recommendations

Based on the findings of this study, this section presents design recommendations applicable to future cultural learning environments. These are organized into four key perspectives.

6.5.1. Supporting the meaning-making of cultural values through experience and dialogue

To enable learners to interpret cultural objects and engage in personal meaning-making, it is crucial to move beyond the passive presentation of knowledge. Instead, experiential and dialogical structures that connect with learners' emotions and memories are more effective. In this study, scenarios 1 and 2 demonstrated how learners initiated meaning-making based on exploration and appreciation activities. For example, in scenario 1, the flow from exploration to discovery, information presentation, and dialogue encouraged heightened interest in cultural objects and enabled learners to reinterpret them in light of their own memories and values.

The following design elements supported such internal meaning-making processes:

- Embodied engagement (e.g., exploration, selection, observation)
- Culturally symbolic objects (e.g., seasonal ingredients and tableware)
- Structures that elicit meaning-making (e.g., quizzes, role assignments)

These elements enabled learners to construct personal meaning from their own perspectives, rather than merely receiving predefined interpretations.

6.5.2. Interaction design to promote collaborative dialogue and joint attention

To support learners in naturally initiating dialogue while focusing on shared objects, it is essential to incorporate non-verbal interaction and spatial arrangements. In this study, bodily expressions such as avatar pointing, gaze, and head orientation were used to establish an environment where learners could intuitively perceive each other's focus of attention.

Particularly in scenario 1, learners engaged in utterances such as “Do you know this?” or “This feels like spring”, which were preceded or accompanied by shared gaze or gestures. Such exchanges facilitated joint attention and shared meaning-making. Furthermore, the spatial layout of exhibits and panels was designed to promote overlapping lines of sight between learners, contributing to the spontaneous emergence of dialogue.

The collaborative learning observed in these scenes was supported by the following elements:

- Visual cues such as gestures and head orientation
- Spatial layouts and pathways conducive to shared attention
- User interfaces that visualize facial expressions and reactions

These design features facilitated the initiation of dialogue and the mutual sharing of meaning, forming a foundation for deeper interpersonal understanding.

6.5.3. Immersive experience design to enhance intrinsic motivation

To encourage learners' voluntary engagement in learning activities, the learning environment must provide an intrinsically motivating experience. Based on IMI (Intrinsic Motivation Inventory) scores and open-ended responses in this study, sensory elements such as the enjoyment of exploration, visual beauty, and the fun of interaction were shown to be strongly linked with intrinsic motivation.

For instance, comments such as “I enjoyed walking through the autumn landscape” and “The tableware was beautiful and conveyed a sense of season” confirmed that interest in culture was naturally drawn out through immersive experience. The learning process was also enriched by questions such as “What is this?” and “What do you think?”, which encouraged learners to construct meaning from the activity itself.

Design elements that supported intrinsic motivation included:

- Sensory staging that evokes seasons and cultural atmosphere (e.g., sound, visuals, motion)
- Freely selectable exploration routes and task sequences that allow learners to explore at their own pace
- Process-oriented task design that promotes discovery and dialogue (e.g., “Which season does this tableware suit?”)

Additionally, the opportunity for learners to “express what they felt in their own words” strongly contributed to deeper engagement. This aligns with the satisfaction of autonomy, relatedness, and competence described in Self-Determination Theory [55] and confirms the effectiveness of VR in supporting motivation.

6.5.4. Scenario and task design to promote reflection

To achieve a deeper understanding of cultural values, it is essential to provide learners with opportunities to reflect on their experiences, connect them with personal memories and emotions, and make meaning from them. Following the reflection model proposed by Boud et al. [15], this study designed scenario structures that naturally induce “returning to the experience”, “attending to feelings”, and “re-evaluating the experience”.

For example, utterances such as “We eat this at home too” or “Chestnut rice reminds me of autumn” linked cultural objects with personal memories. Similarly, remarks like “This tableware has cherry

blossoms—it feels like spring” triggered considerations of cultural background based on intuitive impressions.

Supporting such reflection requires not only delivering information, but also designing scenarios that make learners want to ask questions and re-evaluate their experiences. Narrative and emotionally engaging scenarios tend to be more memorable and thus serve as materials for later reflection. Tasks such as quizzes, questions, and reporting activities serve as scaffolds that help learners reconstruct and verbalize their experiences.

Design elements that promote reflection include:

- Landscapes that evoke a sense of seasonality and everyday life (e.g., autumn leaves, ambient atmosphere).
- Symbolic tableware and ingredients that elicit emotional and personally autobiographical memories.
- Introductory elements that stimulate thought through questions or information prompts.
- Structuring moments of collaborative exploration and reporting to enable shared reflection.

As Jiang et al. [48] have pointed out, in VR environments, it is important not only to design for immersion, but also to intentionally create “downtime for reflection”. In our study as well, we observed learners verbalizing their experiences and reconstructing cultural meanings from a personal perspective. This indicates that the structural design of the scenario and tasks played a role in deepening reflection.

6.6. Limitations and future studies

6.6.1. Scope of cultural dimensions

This study proposes a design framework that supports the interpretation of cultural values grounded in personal experiences and individual sensibilities, focusing on three core elements: ingredients, tableware, and cuisine. As such, the scope of cultural dimensions explored is limited, and does not address broader aspects of culture such as social norms and ceremonial contexts.

In future extensions, incorporating themes that encompass social and ceremonial contexts of food could further enhance learners’ understanding of the diversity and community-based nature of cultural values.

For example, exploring values embedded in behavioral norms such as table manners, order of serving, or with whom and how one shares a meal, as well as ceremonial foods associated with New Year, Obon, and seasonal festivals, could provide learners with opportunities to grasp shared cultural structures that go beyond individual impressions.

These sociocultural themes can also naturally elicit collaborative actions involving role divisions and interpersonal behaviors. For instance, scenarios involving actions such as serving tea, offering the first bite to elders, or preparing festive meals together, can foster learners’ ability to take the perspectives of others and interpret cultural meanings accordingly.

In this way, by addressing the multi-layered nature of cultural values – including sensibility, customs, norms, rituals, and community connections – VR-based cultural learning can move beyond the mere understanding of factual knowledge and facilitate a multi-dimensional awareness and attitudinal formation regarding culture.

6.6.2. Participant backgrounds and VR familiarity

Individual factors such as participants’ backgrounds and familiarity with VR should be considered when interpreting the study’s results. All participants were university students with relatively homogeneous attributes in terms of cultural background, age, and area of specialization. Therefore, there may have been certain tendencies in how cultural values were received and interpreted. While most participants had some experience using VR head-mounted displays, their proficiency

varied. Although those less familiar with VR may have experienced a cognitive or physical burden when navigating or interacting within the virtual space, no significant behavioral differences were observed based on observation analysis. Future studies should collect more detailed data on participants’ cultural backgrounds and VR literacy to enable comparative analyses and the design of supportive training. This may lead to more generalizable findings.

6.7. Transferability to other cultures

In this study, we developed and evaluated a collaborative VR learning environment based on Japanese food culture. As a future direction, applying this framework to other cultures has the potential to establish a more generalizable design model for VR-based cultural learning.

In particular, the design principles derived from this study – scenario-based exploratory tasks, the formation of joint attention through nonverbal interactions, meaning reconstruction through dialogue, and sensory experiences that promote reflection – provide a structural foundation that can be adapted to various cultural contexts.

For instance, in this study, culturally embedded values such as seasonality, the aesthetics of tableware, and cooking processes were presented as objects of sensory and embodied exploration. Learners experienced a process in which they discovered cultural values spontaneously and reinterpreted them through dialogue. These processes of spontaneous awareness of cultural values and reconstruction of meaning through dialogue can serve as effective learning frameworks in other cultural settings as well.

However, the forms of cultural expression, the structures of meaning, and the kinds of experiences that contribute to the emergence of meaning vary significantly across cultures. For example, while Japanese culture emphasizes harmony with nature and the passage of seasons, other cultures may focus more on religious rituals, community ties, or historical memory. Therefore, it is necessary to redesign the scenarios, choose symbolic elements, and construct tasks that elicit engagement appropriate to the target culture.

Drawing on the findings of this study, we identify the following three key perspectives for designing learning environments that can be extended to other cultural contexts:

Experience design that elicits dialogue leading to awareness of cultural values. Rather than unilaterally presenting cultural knowledge, it is effective to support learners in engaging in meaning-making through experience and dialogue. However, the content and significance of the experiences that serve as the starting point for meaning-making differ across cultures, so a deep understanding of the target culture is essential for designing such experiences.

Promotion of dialogue and understanding through nonverbal communication. Nonverbal expressions such as pointing, nodding, and head orientation can induce joint attention and naturally trigger dialogue, even across language barriers. These types of nonverbal interactions are expected to be highly effective in multilingual and multicultural learning environments.

Design of spatial and narrative elements that elicit reflection. In this study, natural landscapes and traditional visual cues contributed to cultural awareness by evoking emotion and memory, rather than relying on excessive information. Such psychological spaces for reflection are also applicable across cultures.

In addition, when applying this framework to other cultures, the learners’ cultural backgrounds and prior understanding of foreign cultures must be taken into account. In intercultural learning settings, learners may encounter unfamiliar practices or values and begin to question why such practices exist or how they differ from their own culture. These questions can serve as catalysts for re-evaluating cultural meaning and may lead to deeper reflection and cultural learning.

However, such awareness does not emerge spontaneously. In order for learners to focus on cultural differences and construct meaning from

them, appropriate instructional support is necessary. Supplementary explanations about cultural contexts, prompts that elicit dialogue, and post-experience reflection guides can all be effective. These strategies help learners go beyond surface-level observations and use cultural differences as opportunities for deeper understanding.

Examples of how this framework could be adapted to other cultural contexts include the following:

- **Islamic cultures:** Meals during Ramadan have strong religious and social significance. In VR scenarios, it would be effective to recreate contexts such as the Iftar meal after sunset, fasting, and prayer rituals [56].
- **Chinese culture:** Dishes like dumplings and fish served during the Lunar New Year, as well as round table dining arrangements, symbolize family reunion and harmony. Recreating these elements in VR could promote deeper understanding of shared cultural values [57].
- **Indian culture:** Eating with one's hands is closely tied to cultural identity. Designing scenarios that explain this practice and encourage reflective dialogue would help learners better understand the meaning behind such embodied customs [58].

While the design knowledge presented in this study shows promise for cross-cultural application, it must be adapted flexibly to the specific symbolic systems, cultural contexts, and learner backgrounds of the target cultures. Future work should involve comparative studies and empirical validation with learners from diverse cultural backgrounds to establish VR-based cultural learning environments that balance generality with adaptability.

7. Conclusion

This study aimed to explore how collaborative learning can support learners in interpreting and understanding cultural values through virtual experiences. To this end, we developed a scenario-based VR learning environment focused on Japanese food culture. In the system, learners engaged in collaborative activities using avatars, sharing a series of cultural experiences such as exploration, observation, and cooking. Through these interactions, they verbalized and empathized with each other's perspectives and insights, deepening their understanding of cultural values.

The results of the comparative experiment showed that learners in the collaborative learning condition demonstrated significantly greater gains than those in the individual learning condition in terms of cultural value recognition, knowledge acquisition, intrinsic motivation, and cultural awareness in daily life. Additionally, analysis of observational data and open-ended responses revealed that nonverbal cues such as pointing gestures, nodding, and head orientation helped establish joint attention and facilitated natural conversation and interaction. Moreover, it was observed that within these dialogues, learners naturally engaged in reflection, reconstructing cultural meaning while revisiting their own experiences and emotions.

Based on these findings, we proposed four design recommendations for supporting cultural value interpretation in virtual learning environments: (1) supporting meaning-making through experience and dialogue, (2) supporting collaborative interaction, (3) designing immersive environments to promote intrinsic motivation, and (4) creating scenarios and tasks that encourage reflection. In particular, the study highlighted the importance of embedding cultural values within the learning experience itself, rather than teaching them explicitly, and of supporting a learning process in which learners share and reconstruct these values with others.

As future directions, we propose incorporating more socially and ceremonially embedded cultural contexts (such as etiquette and annual events) to deepen the understanding of diverse cultural values rooted in social and communal practices.

In addition, by applying the design insights gained in this study to different cultural regions, further investigations are needed to examine the generality and adaptability of VR-based cultural learning environments. Since cultural meanings differ greatly across countries and regions – for example, in religious practices, dietary customs, and etiquette – scenario design and methods for eliciting dialogue must be redesigned according to each cultural context. Future work should empirically explore whether collaborative, experiential, and reflective support structures in VR learning can be adapted across different cultural backgrounds through a multicultural perspective.

CRedit authorship contribution statement

Boning Li: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lehao Zhou:** Investigation. **Junichi Hoshino:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

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During the preparation of this work, the authors used ChatGPT and Grammarly in order to translate and improve readability. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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